

Lecture 4: Linear Regression

ML/AI for Engineering Applications
ENV 6932

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Fall 2025

Outline

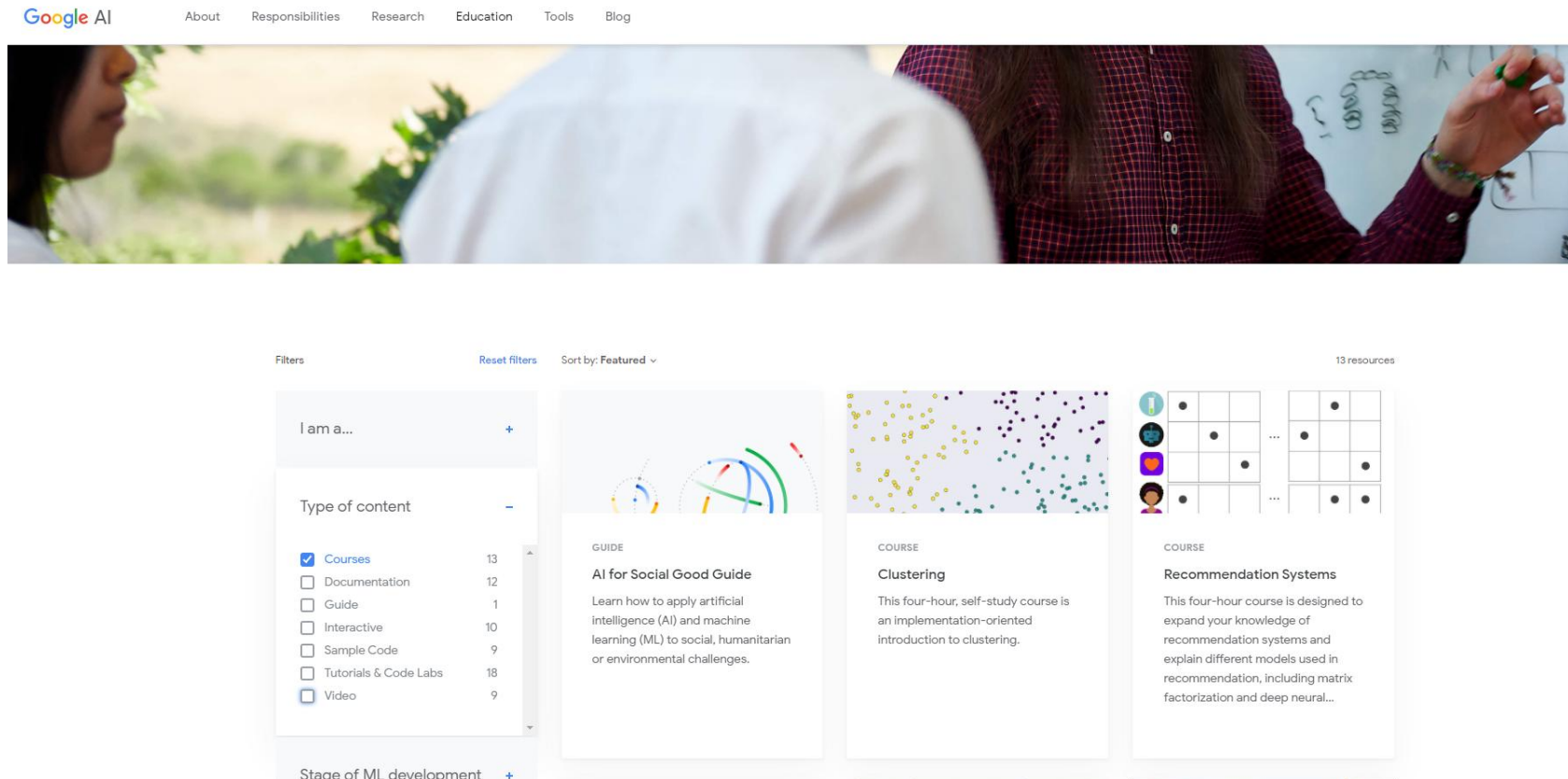
- Guidelines on term paper
- What is regression?
- Simple linear regression
- Multiple linear regression
- Model validation

About term paper

- Make yourself familiar with Google AI
- Install Anaconda

About term paper

- Make yourself familiar with Google AI



<https://ai.google/education/>

Data sources for term paper

- Google AI > Using AI for social good > Data Sources
- [Dataset Search](#)
- [Google Earth Engine Catalog](#)
- [Kaggle](#)

The Kaggle logo, featuring the word "kaggle" in a lowercase, blue, sans-serif font.

Kaggle Datasets

Explore, execute, share, and comment on code for any open dataset with the in-browser analytics tool, Kaggle Kernels. You can also download datasets in an easy-to-read format.



Google Earth Engine Catalog

Earth Engine's public data archive includes more than forty years of historical imagery and Earth science datasets, updated and expanded daily.

The Google Dataset Search interface, showing the Google logo, the text "Dataset Search", and a search bar.

Dataset Search

Google's Dataset Search tool allows users to search across thousands of dataset repositories on the Web. Once you've found a relevant dataset, Dataset Search will direct you to the repository ...

Teachable Machine

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How do I use it?

Class 1

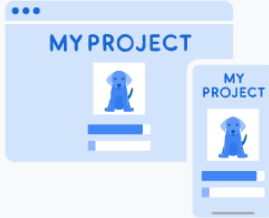


Class 2



TRAIN MODEL





- 1 Gather**

Gather and group your examples into classes, or categories, that you want the computer to learn.

[Video: Gather samples](#) 
- 2 Train**

Train your model, then instantly test it out to see whether it can correctly classify new examples.

[Video: Train your model](#) 
- 3 Export**

Export your model for your projects: sites, apps, and more. You can download your model or host it online for free.

[Video: Export your model](#) 

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Individual Edition

Your data science toolkit

With over 25 million users worldwide, the open-source Individual Edition (Distribution) is the easiest way to perform Python/R data science and machine learning on a single machine. Developed for solo practitioners, it is the toolkit that equips you to work with thousands of open-source packages and libraries.

Anaconda Individual Edition

Download 

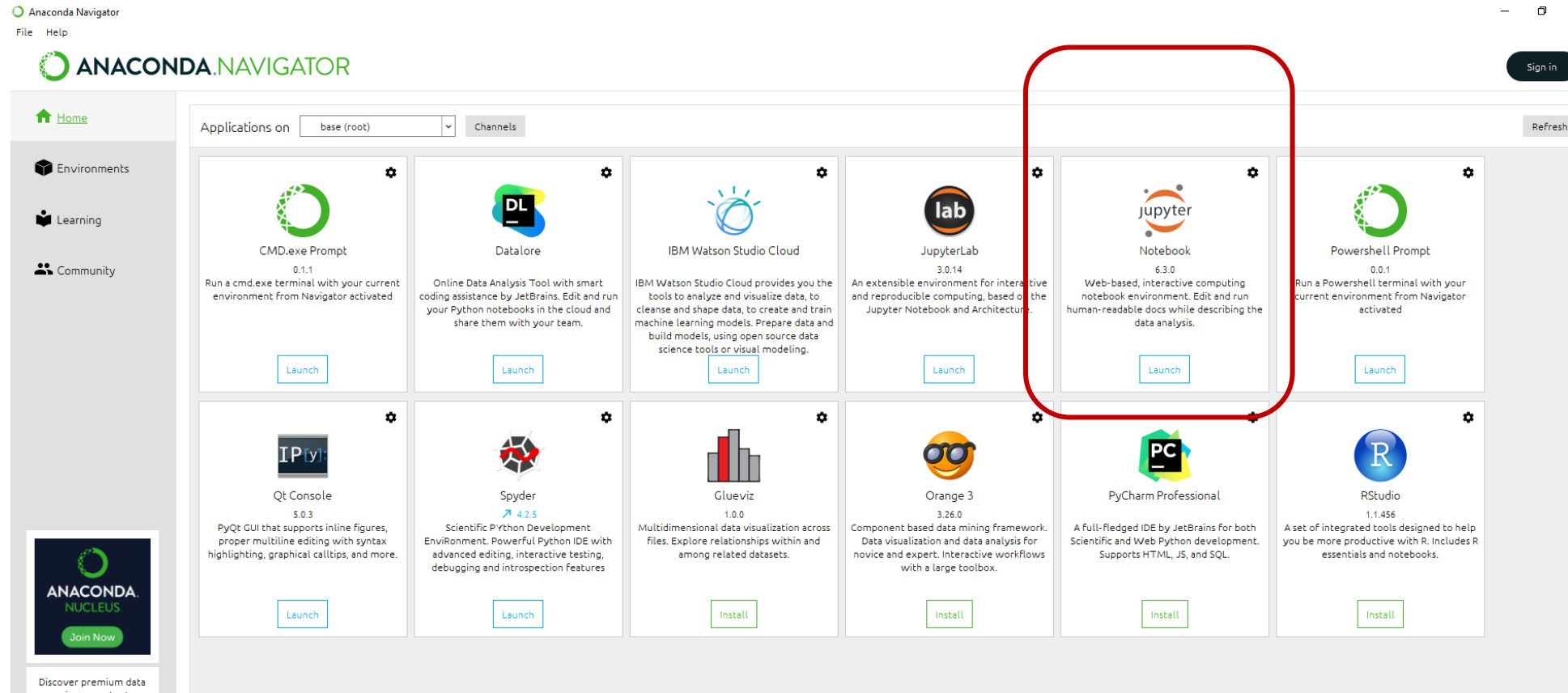
For Windows

Python 3.8 • 64-Bit Graphical Installer • 477 MB

Get Additional Installers



Jupyter notebook



What is Regression?

- Regression is the process of predicting a continuous value.
- **Regression:** reduction of the complexity of a system into a simpler form.
- When to use regression methods? to predict a continuous values such as Nitrogen Dioxide (NO₂), using some other variables
 - Temperature in °C
 - Relative Humidity (%)
 - Absolute Humidity

Example: Air Quality

Air Quality Data Set

Download: [Data Folder](#), [Data Set Description](#)

Abstract: Contains the responses of a gas multisensor device deployed on the field in an Italian city. Hourly responses averages are recorder

Data Set Characteristics:	Multivariate, Time-Series	Number of Instances:	9358	Area:	Computer
Attribute Characteristics:	Real	Number of Attributes:	15	Date Donated	2016-03-23
Associated Tasks:	Regression	Missing Values?	Yes	Number of Web Hits:	559481

Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)	PT08.S5(O3)	T	RH	AH
3/10/2004	18:00:00	2.6	1360	150	11.9	1046	166	1056	113	1692	1268	13.6	48.9	0.7578
3/10/2004	19:00:00	2	1292	112	9.4	955	103	1174	92	1559	972	13.3	47.7	0.7255
3/10/2004	20:00:00	2.2	1402	88	9.0	939	131	1140	114	1555	1074	11.9	54.0	0.7502
3/10/2004	21:00:00	2.2	1376	80	9.2	948	172	1092	122	1584	1203	11.0	60.0	0.7867
3/10/2004	22:00:00	1.6	1272	51	6.5	836	131	1205	116	1490	1110	11.2	59.6	0.7888
3/10/2004	23:00:00	1.2	1197	38	4.7	750	89	1337	96	1393	949	11.2	59.2	0.7848
3/11/2004	0:00:00	1.2	1185	31	3.6	690	62	1462	77	1333	733	11.3	56.8	0.7603
3/11/2004	1:00:00	1	1136	31	3.3	672	62	1453	76	1333	730	10.7	60.0	0.7702
3/11/2004	2:00:00	0.9	1094	24	2.3	609	45	1579	60	1276	620	10.7	59.7	0.7648
3/11/2004	3:00:00	0.6	1010	19	1.7	561	-200	1705	-200	1235	501	10.3	60.2	0.7517
3/11/2004	4:00:00	-200	1011	14	1.3	527	21	1818	34	1197	445	10.1	60.5	0.7465
3/11/2004	5:00:00	0.7	1066	8	1.1	512	16	1918	28	1182	422	11.0	56.2	0.7366
3/11/2004	6:00:00	0.7	1052	16	1.6	553	34	1738	48	1221	472	10.5	58.1	0.7353
3/11/2004	7:00:00	1.1	1144	29	3.2	667	98	1490	82	1339	730	10.2	59.6	0.7417
3/11/2004	8:00:00	2	1333	64	8.0	900	174	1136	112	1517	1102	10.8	57.4	0.7408
3/11/2004	9:00:00	2.2	1351	87	9.5	960	129	1079	101	1583	1028	10.5	60.6	0.7691
3/11/2004	10:00:00	1.7	1233	77	6.3	827	112	1218	98	1446	860	10.8	58.4	0.7552
3/11/2004	11:00:00	1.5	1179	43	5.0	762	95	1328	92	1362	671	10.5	57.9	0.7352
3/11/2004	12:00:00	1.6	1236	61	5.2	774	104	1301	95	1401	664	9.5	66.8	0.7951
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3/11/2004	14:00:00	2.9	1371	164	11.5	1034	207	983	128	1730	1037	8.0	81.1	0.8736
3/11/2004	15:00:00	2.2	1310	79	8.8	933	184	1082	126	1647	946	8.3	79.8	0.8778
3/11/2004	16:00:00	2.2	1292	95	8.3	912	193	1103	131	1591	957	9.7	71.2	0.8569
3/11/2004	17:00:00	2.9	1383	150	11.2	1020	243	1008	135	1719	1104	9.8	67.6	0.8185
3/11/2004	18:00:00	4.8	1581	307	20.8	1319	281	799	151	2083	1409	10.3	64.2	0.8065
3/11/2004	19:00:00	6.9	1776	461	27.4	1488	383	702	172	2333	1704	9.7	69.3	0.8319

Example: Benzene estimation in urban pollution monitoring

- **Attribute Information:**

0 Date (DD/MM/YYYY)

1 Time (HH.MM.SS)

2 True hourly averaged concentration CO in mg/m^3

3 PT08.S1 (tin oxide) hourly averaged sensor response

4 True hourly averaged overall Non Metanic HydroCarbons concentration in microg/m^3

5 True hourly averaged **Benzene concentration** in microg/m^3

6 PT08.S2 (titania) hourly averaged sensor response

7 True hourly averaged NOx concentration in ppb

8 PT08.S3 (tungsten oxide) hourly averaged sensor response

9 True hourly averaged NO2 concentration in microg/m^3

10 PT08.S4 (tungsten oxide) hourly averaged sensor response

11 PT08.S5 (indium oxide) hourly averaged sensor response

12 Temperature in $^{\circ}\text{C}$

13 Relative Humidity (%)

14 AH Absolute Humidity

Regression Model

- **The dependent variable:** the state, target, or final goal
- **Independent variables:** also known as explanatory variables, can be viewed as the causes of those states.
- **The independent variables are shown conventionally by X,** and the dependent variable is notated by Y.
- Our regression model relates Y, to a function of X.
- $Y = f(X)$

Key Point

- The dependent variable (Y) should be continuous and cannot be a discrete value. However, the independent variable or variables (X), can be measured on either a categorical, or continuous scale.

Types of regression models

- Simple regression: **one independent variable is used to estimate a dependent variable**
 - Linear regression
 - Non-linear regression
- Multiple regression: **more than one independent variables**
 - Multiple linear regression
 - Multiple non-linear regression

Simple Regression

- A single x is used to predict y

$$y = \beta_0 + \beta_1 x_1$$

- β_0 and β_1 : the parameters of the line that we must adjust.
 - β_1 : the slope or gradient of the fitting line
 - β_0 : the intercept
- β_0 and β_1 : also called the coefficients of the linear equation.
- **You can interpret this equation as $\hat{y}$ being**

$$\hat{y} = \beta_0 + \beta_1 x_1$$

The General Model for Regression

Let's define:

- Regression analysis identifies the function f such that:

$$y_i = f(\beta, X) + e_i$$

- i : which refers to the i – th observation in a dataset
- β : a set of parameters for the model
- e : the random error associated with a given observation

Limits of Regression

- Regression can tell us about correlation, but not causality.
- **A good regression model:** maybe the variables that we study are causally related to one another.
- **Not a well-fitting model:** normally assume that no causal relationship exists between them.
- However, the literature is full of examples of variables that were believed to be causally related but weren't, and vice versa.

Model Evaluation

- The actual function:

$$y_i = f(\beta, X) + e_i$$

- The predicted function:

$$\hat{y}_i = f(\beta, X)$$

- The error:

$$e_i = y_i - \hat{y}_i$$

Finding the model parameters

- The typical error metric: the sum of the squared errors
- $E = \sum_{i=1}^n e_i^2$
- The problem of identifying the linear regression model for two variables x, y can thus be reformulated as the finding β_1 and β_0 which minimize the sum of squared errors.
- $\min(a, b) : E(a, b) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \beta_1 x_i - \beta_0)^2$

Computing the Solution to Linear Regression

$$y_i \approx \beta_0 + \beta_1 x_i$$

β_0 = intercept

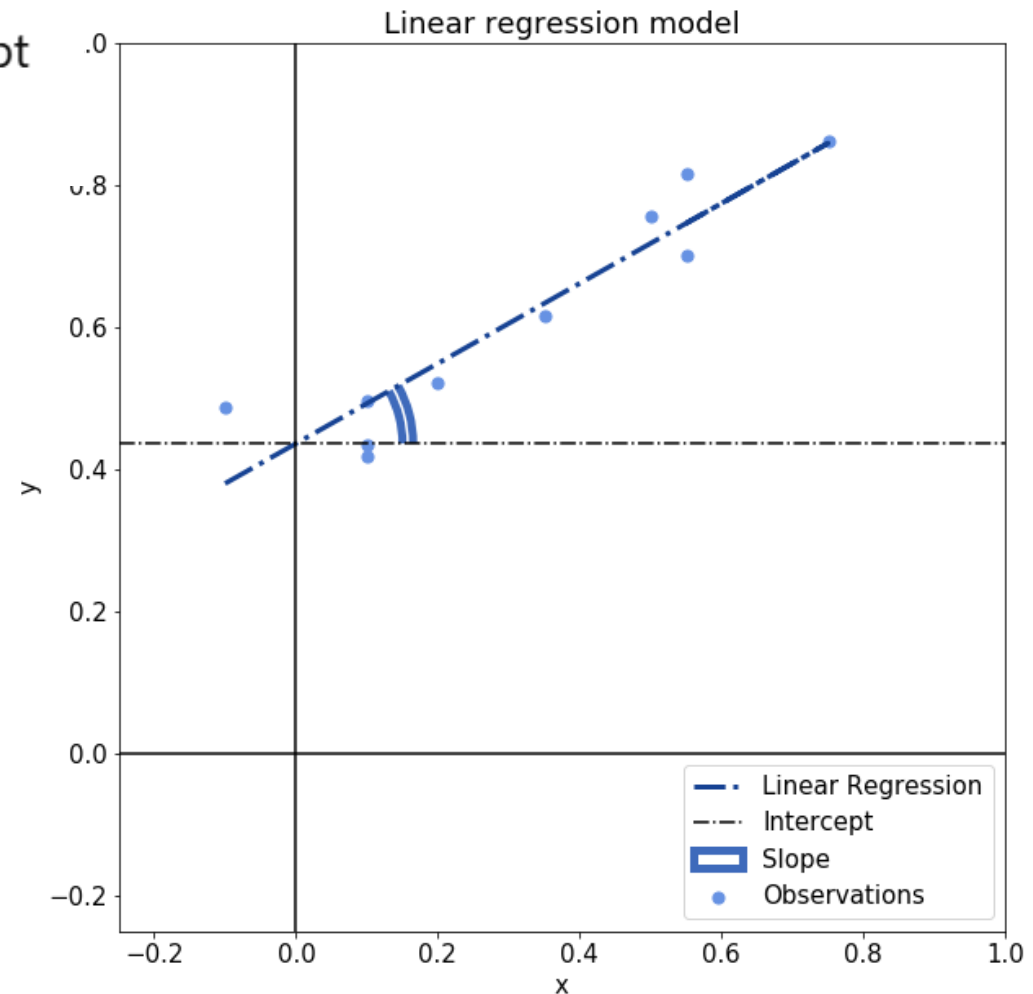
β_1 = slope

$$e_i = y_i - (\beta_0 + \beta_1 x_i)$$

$$MSE(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

$$\frac{\partial MSE}{\partial \beta_0} = -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

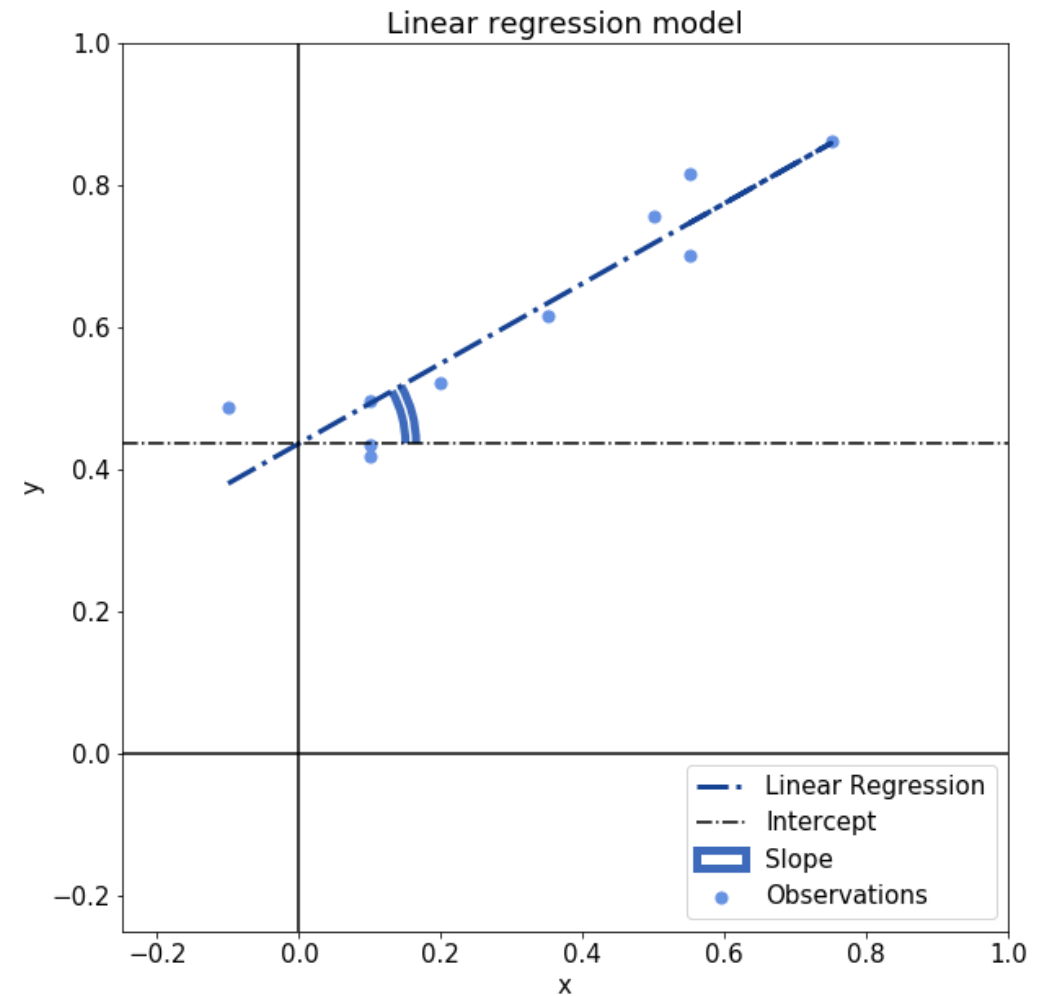
$$\frac{\partial MSE}{\partial \beta_1} = -\frac{2}{n} \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$



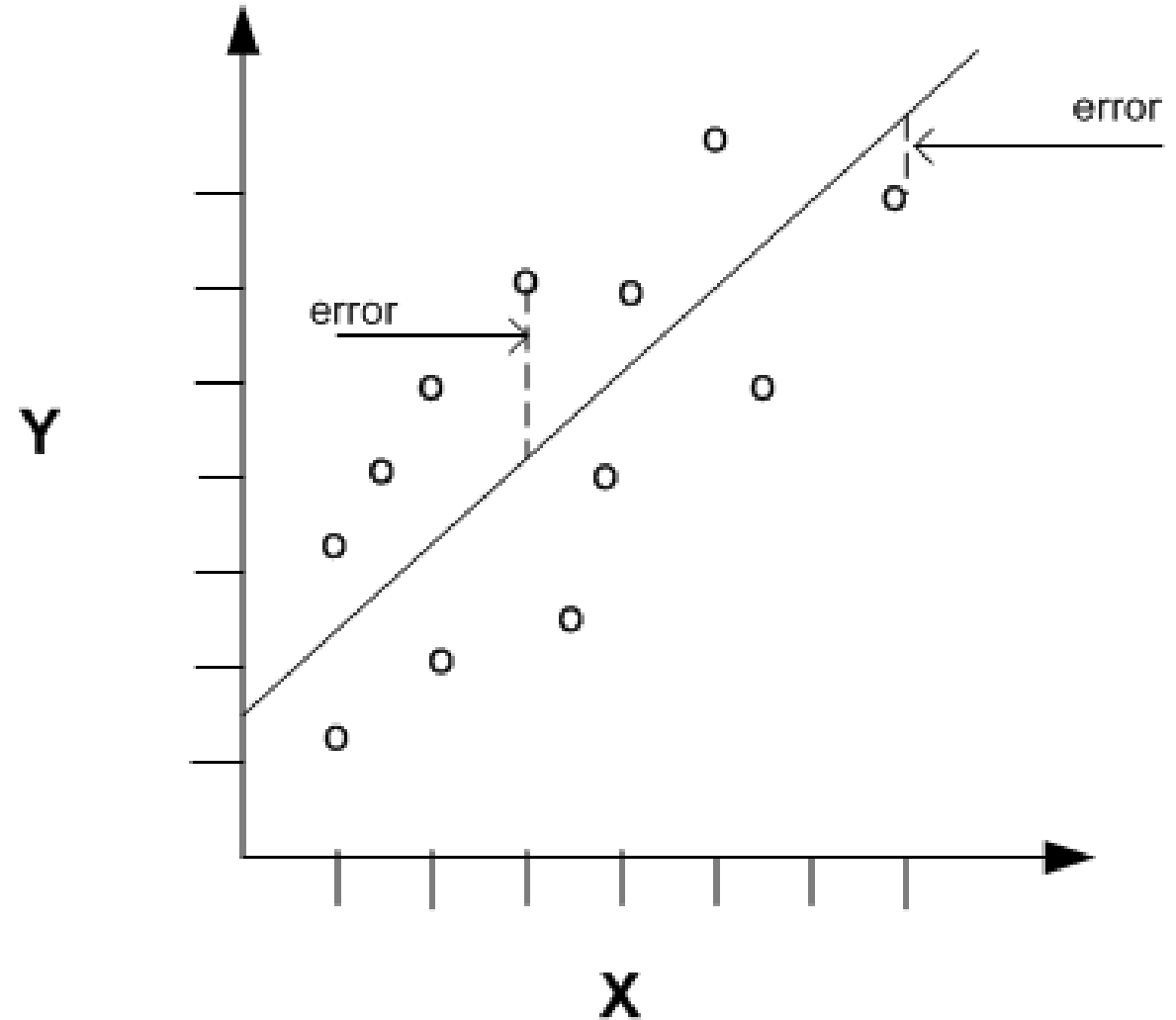
Computing the Solution to Linear Regression

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$



Model evaluation



Model Evaluation

- The goal of regression: to build a model to accurately predict an unknown case.
- We will perform regression evaluation after building the model.
- Two types of evaluation techniques:
 - (1) Train and test on the same dataset**
 - (2) Train/test split**

Model Evaluation

- **Question:** how much can we trust this model for prediction?
- **One solution:** select a portion of our dataset for testing.
- We use the entire dataset for training, and we build a model using this training set. Now, we select a small portion of the dataset, as a test set.

Train and Test on the same dataset

- This evaluation technique would most likely have:
 - high training accuracy and
 - low out-of-sample accuracy
- Since the model knows all of the testing data points from the training.
- Training accuracy: the percentage of correct predictions that the model makes when using the test dataset.
- Out-of-sample accuracy: the percentage of correct predictions that the model makes on data that the model has not been trained on.
- How to improve out-of-sample accuracy (the main purpose of prediction)?

Train/test split

- More realistic for real-world problems
- More accurate evaluation on out-of-sample accuracy.
- However, make sure to train the model with the testing set afterwards, not to lose potentially valuable data.
- Issue: it is highly dependent on the datasets on which the data was trained and tested.

Evaluation Metrics

- Several evaluation metrics:
 - **Mean Squared Error**
 - **Root Mean Squared Error**
 - **Mean Absolute Error**
 - **Relative Absolute Error, also known as Residual Sum of Squares**
 - **Relative Squared Error**
- Error: the difference between the data points and the trend line generated by the algorithm.

Mean Square Error and Root Mean Square Error

- MSE is an absolute measure of the goodness.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

Mean Absolute Error (MAE)

- Compared to MSE or RMSE, MAE is a more direct representation of sum of error terms.
- MSE gives larger penalization to big prediction error by square it while MAE treats all errors the same.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

R Squared/Adjusted R Square

- R Squared is calculated by the sum of square of prediction error divided by the total sum of the square which replaces the calculated prediction with mean.
- R Squared is a relative measure of how well the model fits dependent variables.

$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Adjusted R-squared

- The main difference between adjusted R-squared and R-squared: R-squared describes the amount of variance of the y represented by every single independent variable x.
- Adjusted R-squared measures variation explained by only the independent variables that actually affect y.

$$R^2_{adjusted} = \left[\frac{(1-R^2)(n-1)}{n-k-1} \right]$$

Summary of Evaluation Metrics:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}|$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2}$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y})^2}{\sum (y_i - \bar{y})^2}$$

The Choice of Metric:

- Each of these metrics can be used for quantifying of your prediction.
- The choice of metric completely depends on the type of model, your data type, and domain of knowledge.

Overfitting and Underfitting a Model

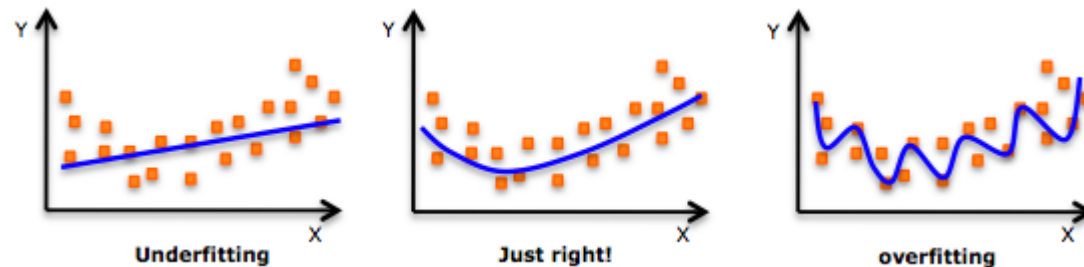
- We usually split our data into two subsets: training data and testing data.
- Sometimes dataset is split into three: train, validate and test.
- We fit our model on the train data, in order to make predictions on the test data.
- One of two things may happen: we overfit our model or we underfit our model.

Overfitting and Underfitting

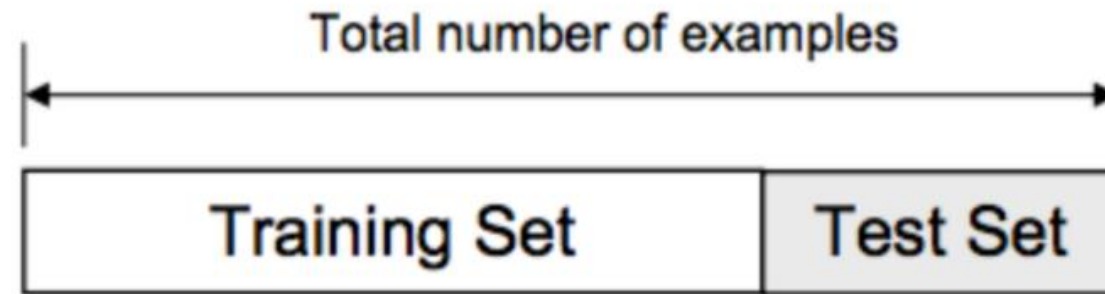
- **Overfitting:** the model is trained “too well” and is now, fit too closely to the training dataset.
- **When this happens?** when the model is too complex, when we have too many features/variables compared to the number of observations
- This model will be very accurate on the training data but not accurate on new data.
- The model is not generalized since the model learns or describes the “noise” in the training data instead of the actual relationships between variables in the data.
- This noise, obviously, will not be present in new dataset, and cannot be applied to it.

Underfitting

- **Underfitting:** means the model does not fit the training data and therefore misses the trends in the data.
- Underfitting is the result of a very simple model (not enough predictors/independent variables). When a linear model is fitted to nonlinear data.
- The model cannot be generalized to new data.



How to avoid overfitting/underfitting?



Caveats in the train/test split method

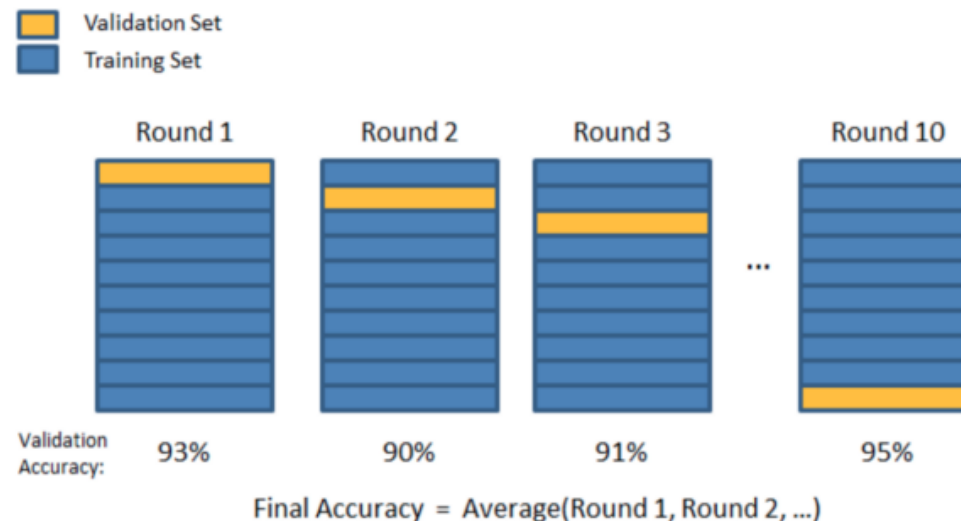
- Train/test split has its dangers
- What if the split isn't random? What if one subset of our data has only observations from a certain state, income level, gender, certain age? This will result in overfitting.
- **Solution:** Cross validation

Cross validation methods

- There are a bunch of cross validation methods.
- **K-Folds Cross Validation**
- **Leave One Out Cross Validation (LOOCV)**
- **Stratified k-Fold Cross Validation**

K-Folds Cross Validation

- Split the entire data into k different subsets (or folds).
- Use k-1 subsets to train the data and leave the last subset (or the last fold) as test data.
- Then average the model against each of the folds

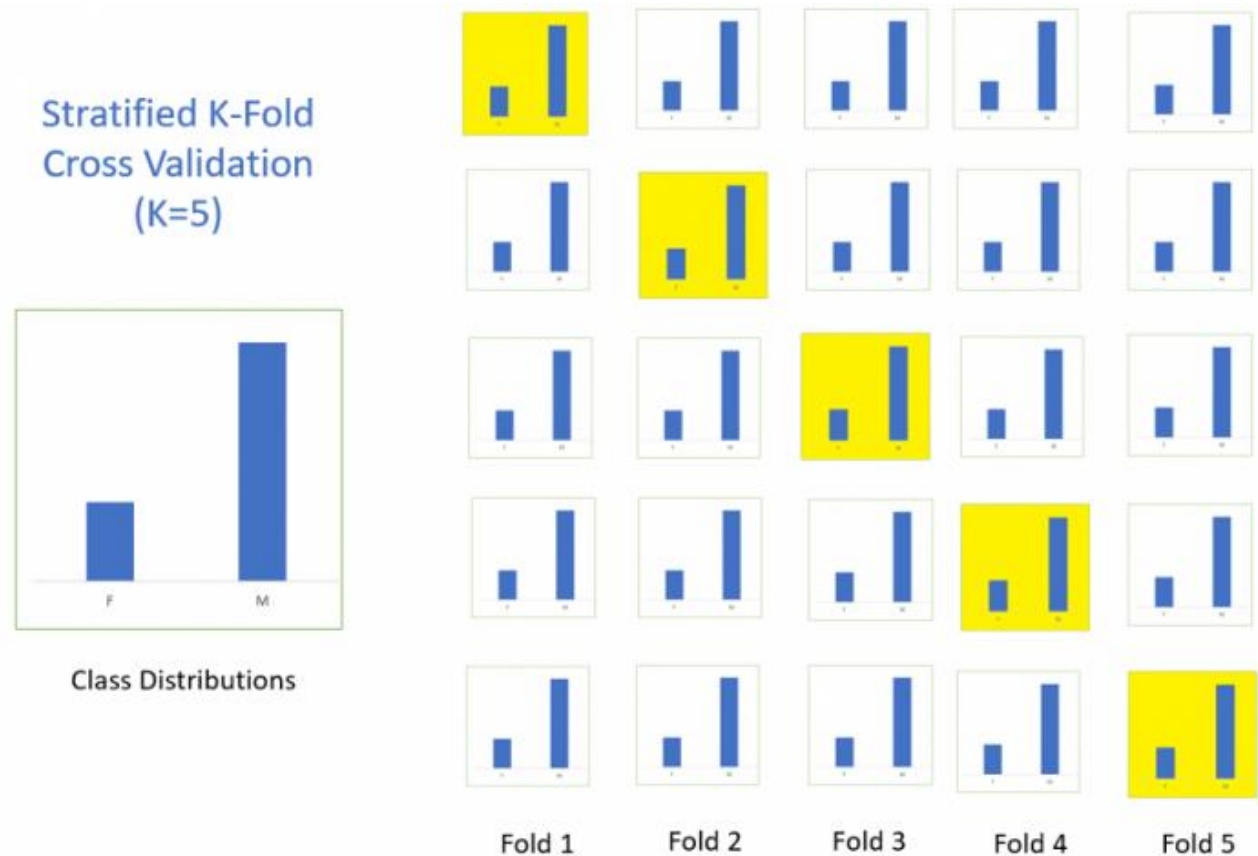


Leave One Out Cross Validation (LOOCV)

- we reserve only one data point from the available dataset and train the model on the rest of the data.
- This process iterates for each data point (process is repeated n times)
- Lower bias since we use all data points
- The estimation is influenced by the data point, an outlier can lead to a higher variation.

Stratified k-Fold Cross Validation

- The splitting of data into folds is governed by criteria such as making sure that each fold has the same proportion of observations with a given categorical value (gender, age group, education level).
- **Example:** the stratified k-fold validation is set on basis of Gender whether M or F



Multiple Linear Regression

- When multiple independent variables are present, the process is called multiple linear regression.
- Extension of simple linear regression
- Two applications for multiple linear regression:
 - First, which independent variables affect the dependent variable.
 - Second, how each independent variable (feature) affects the dependent variable. (the strength of effect)

Multiple Linear Regression

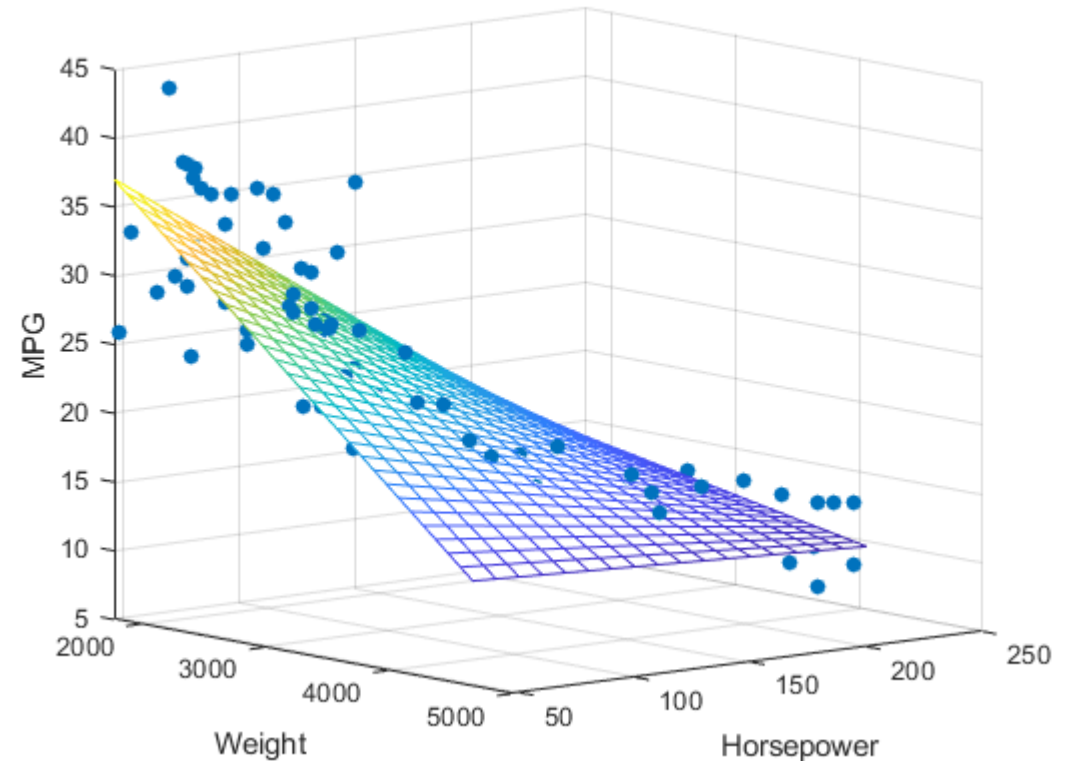
- **Mathematical representation:** Mathematically, we can show it as a dot product of two vectors; the parameters vector and the feature set vector.
- Dependent variable is a linear combination of multiple variables

$$y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \varepsilon$$

Multiple Linear Regression

Example: estimating selling price

- Kilometer information,
- Engine power,
- Year of production,
- Damage rate



<https://medium.com/analytics-vidhya/new-aspects-to-consider-while-moving-from-simple-linear-regression-to-multiple-linear-regression-dad06b3449ff>

How to find parameters of the model, β ?

- Objective of multiple linear regression is to minimize the MSE equation.
- How to estimate β ?
- **Common methods:** the ordinary least squares and optimization approach.
- Ordinary least squares estimates the values of the coefficients by minimizing the mean square error. **It uses the data as a matrix and uses** linear algebra operations to estimate the optimal values for the model parameters β .
- The ordinary least squares is proper for low size datasets (<10k rows)

Multiple Linear Regression

- Optimization
- Gradient decent: That is, you can use a process of optimizing the values of the coefficients by iteratively minimizing the error of the model on your training data.
- For example, you can use gradient descent which **starts optimization with random values for each coefficient**, then calculates the errors and tries to minimize it through changing of the coefficients in multiple iterations.
- Gradient descent is a proper approach if you have a large data set.

Multiple Linear Regression

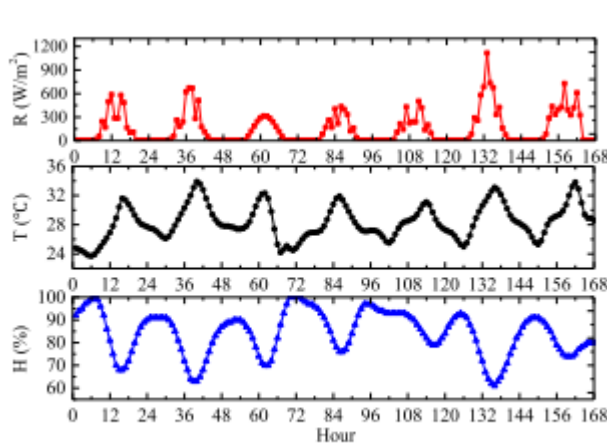
- There are a number of ways to check for linear relationship.
- For example, you can use scatter plots and then visually checked for linearity.
- If data does not show linear trend, then use non-linear regression.

Can independent variables be categorical?

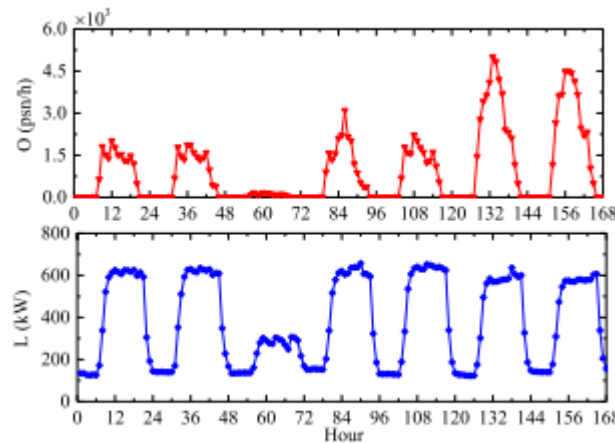
- Yes, but categorical Independent variables should be converted to numerical values.
- If we use too many independent variables, then model may overfit.

Nonlinear Regression

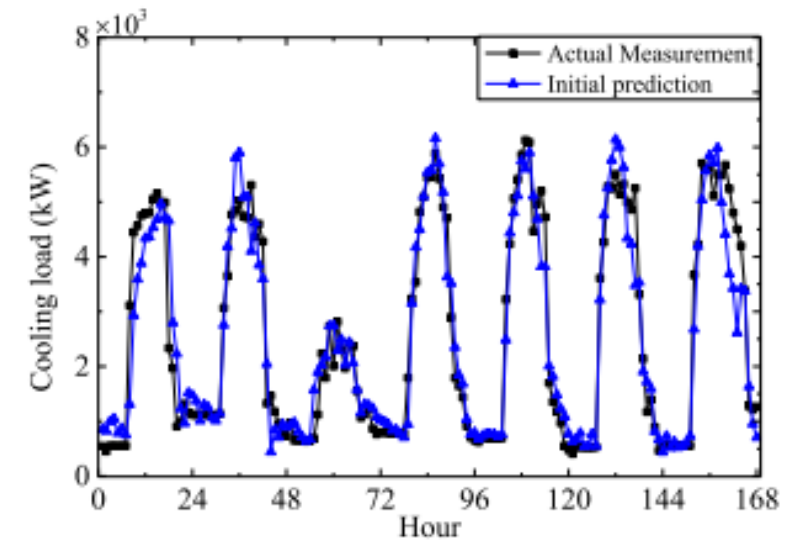
- Example: Cooling load prediction of HVAC systems
- Inputs: temperature (T), relative humidity (H), solar radiation (R), occupancy changes (O), and light power (L)



(a) Outdoor variables



(b) Indoor variables

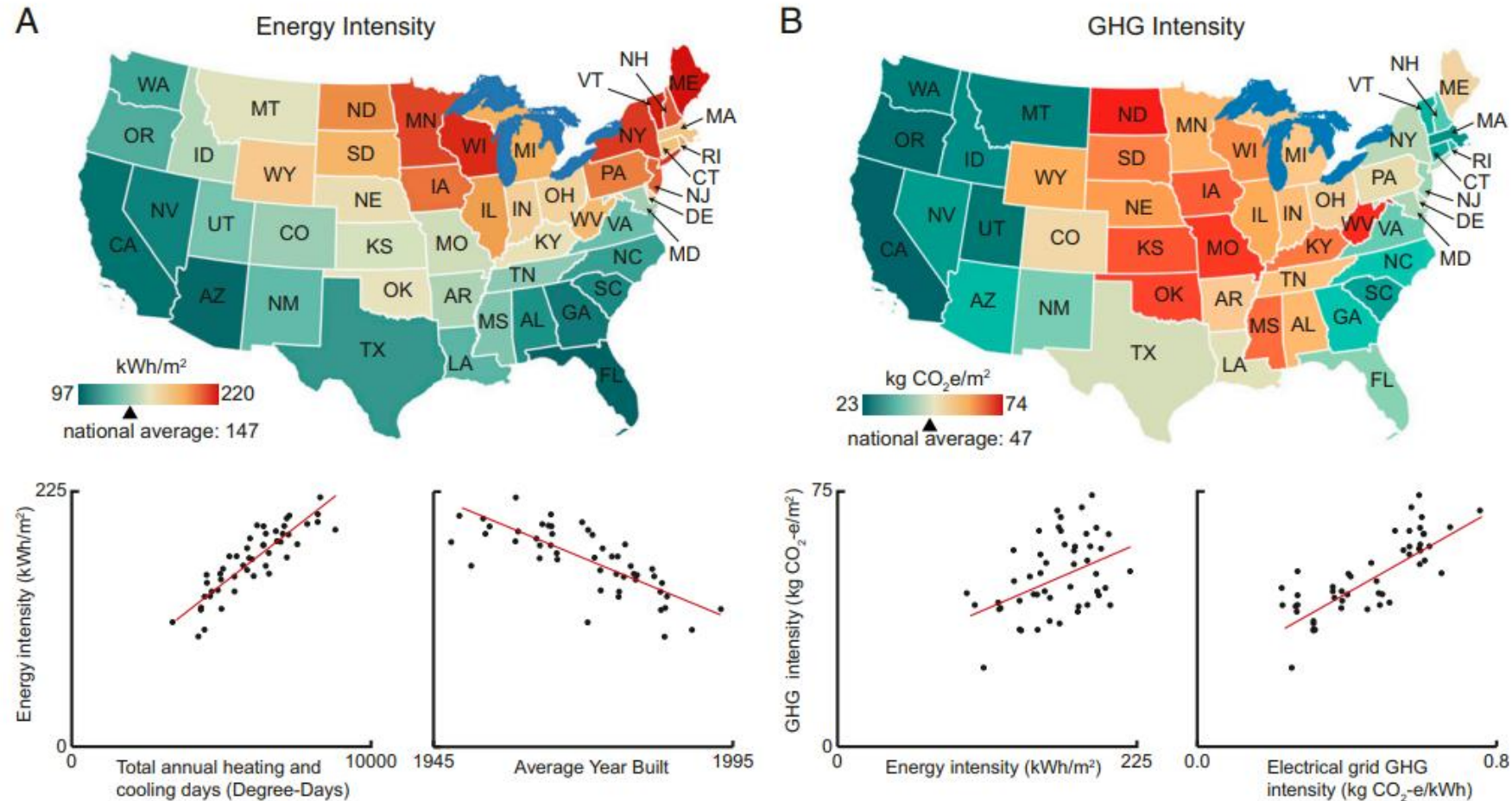


$$\begin{aligned}
 CL(p, t + \tau) = & w_{1,t} \cdot T_{t+\tau} + w_{2,t} \cdot T_{t+\tau}^2 + w_{3,t} \cdot H_{t+\tau} + w_{4,t} \cdot T \cdot H \\
 & + w_{5,t} \cdot R_{t+\tau} + w_{6,t} \cdot O_{t+\tau} + w_{7,t} \cdot L_{t+\tau} \\
 & + w_{8,t} \cdot CL_{a,t} + w_{9,t}
 \end{aligned}$$

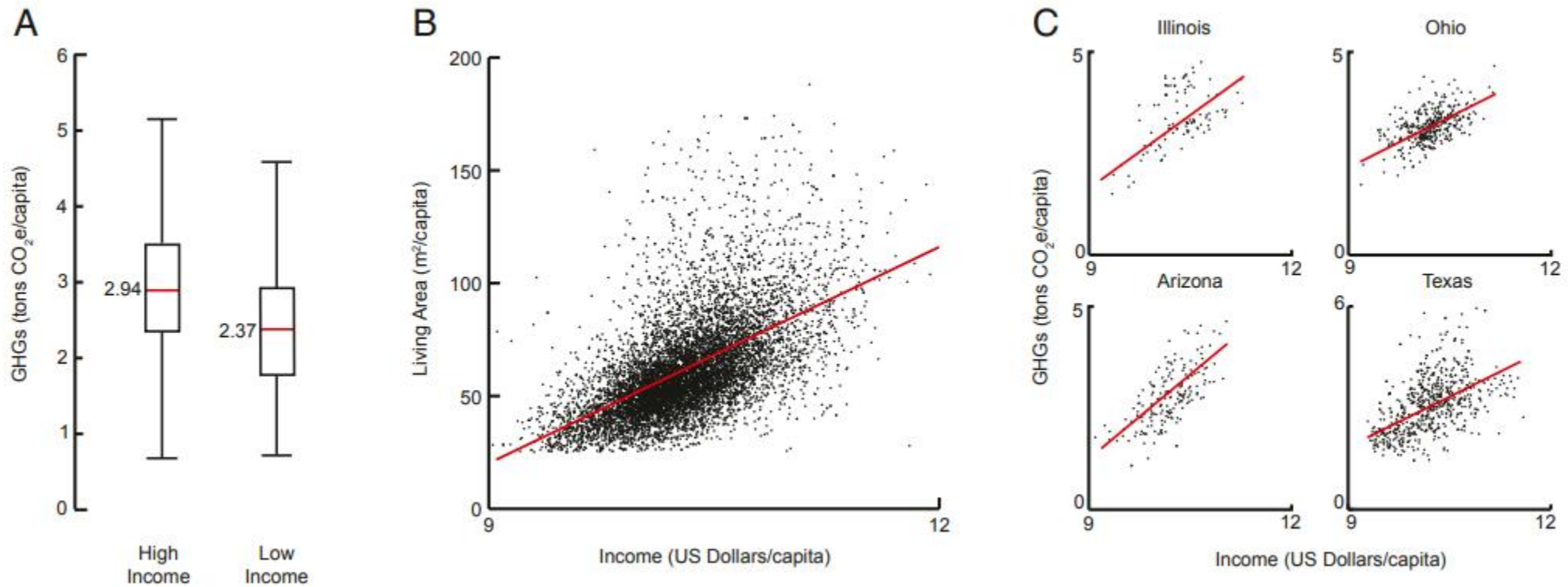
Example: Carbon footprint of household energy use in the US

- Residential energy consumption accounts for roughly 20% of GHG emissions in the USA.
- Using data on 93 million individual households, we estimate these GHGs among the contiguous USA and clarify the impact of climate, affluence, and building attributes (age, housing type, heating fuel) on GHG.

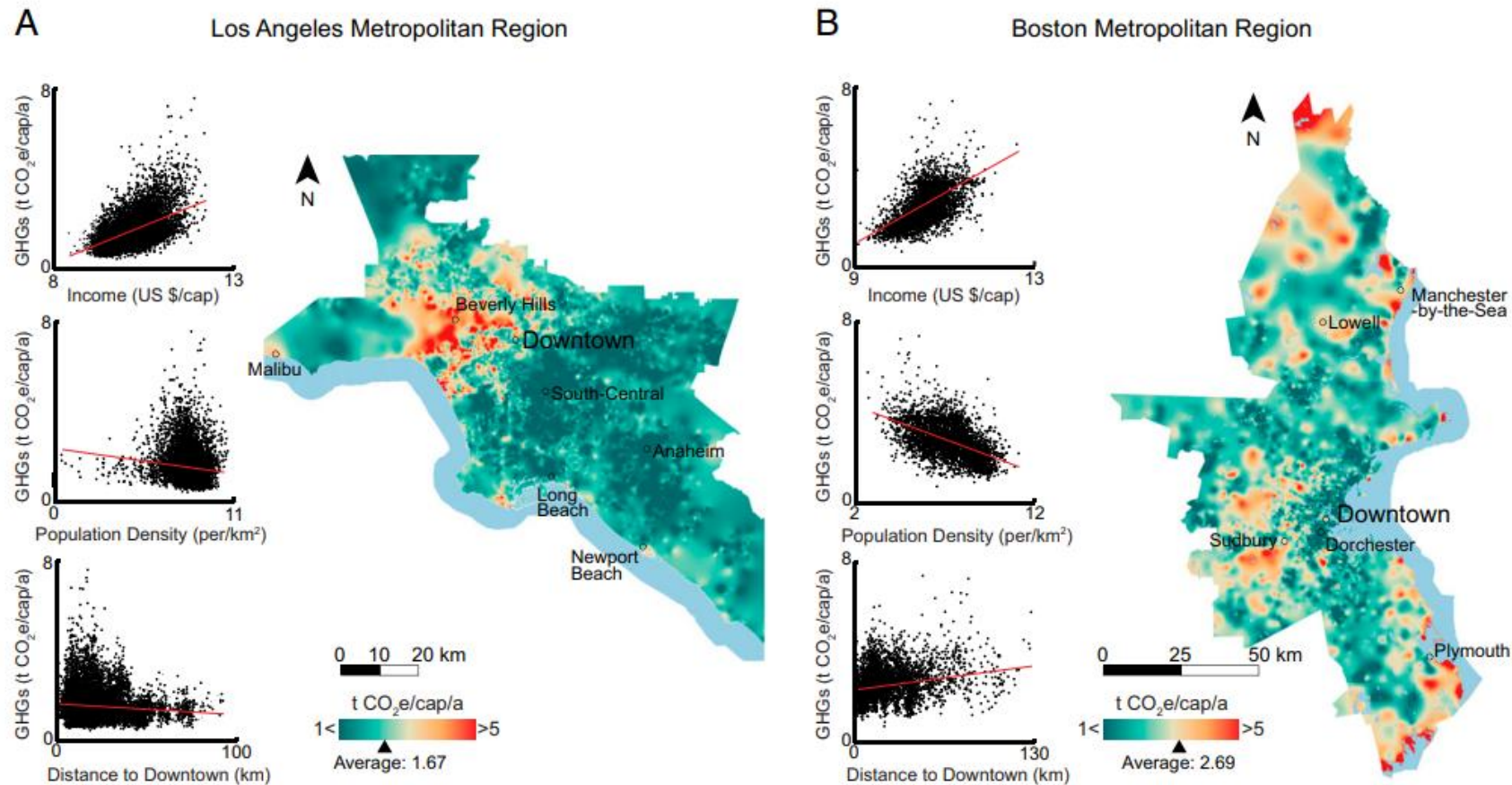
Example: Carbon footprint of household energy use in the US



Example: Carbon footprint of household energy use in the US



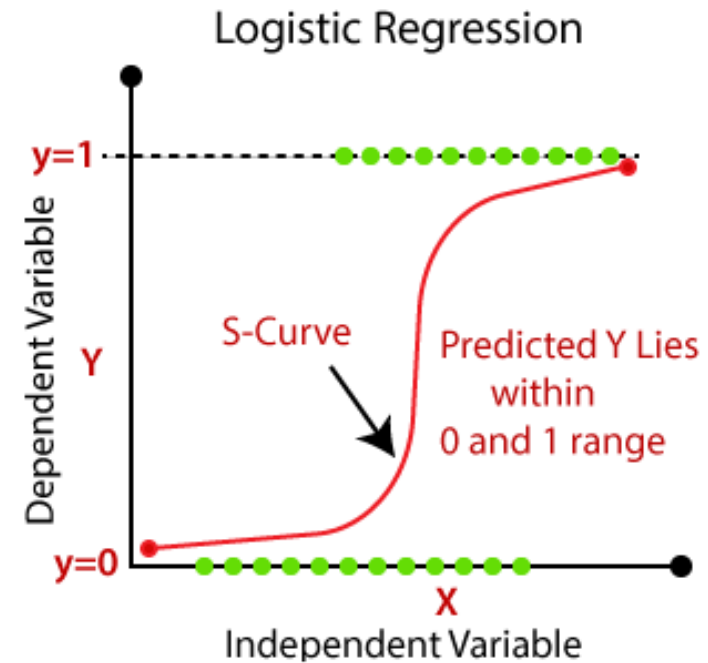
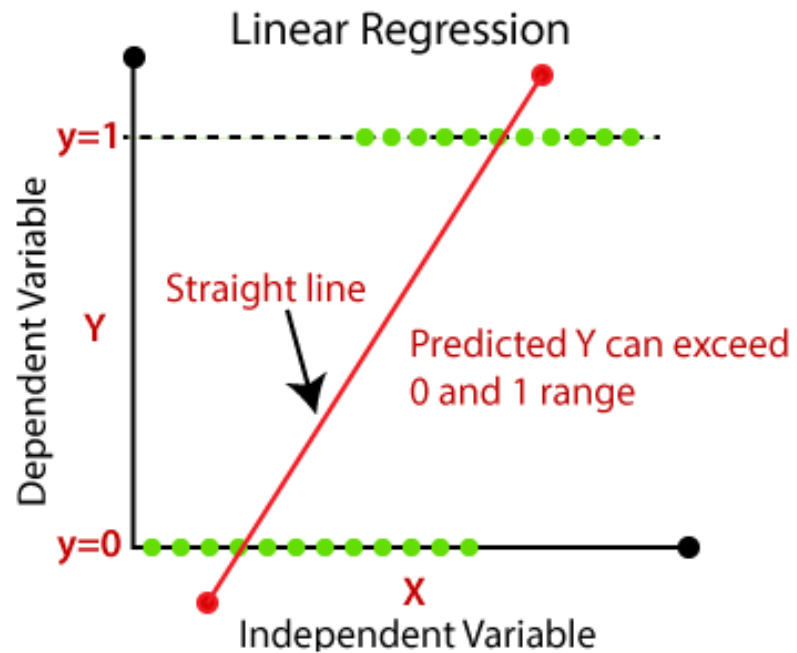
Example: Carbon footprint of household energy use in the US



Findings: Carbon footprint of household energy use in the US

- GHGs (per unit floor space) are lowest in Western US states and highest in Central states.
- Wealthier Americans have per capita ~25% higher footprints than those of lower-income residents, primarily due to larger homes.
- In especially affluent suburbs, these emissions can be 15 times higher than nearby neighborhoods.

Logistic Regression

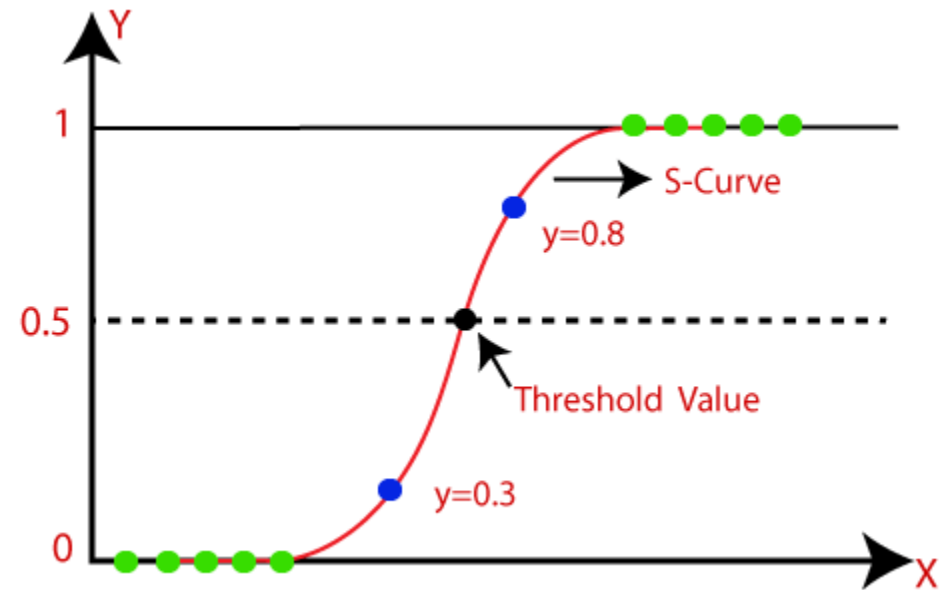


Logistic Regression

- One of the most popular ML algorithm.
- Used for Classification as well as Regression, but mainly Classification.
- The output of Binary Logistic Regression can be only between the 0 and 1.
- **Example:** whether it will rain today or not, 0 or 1, true or false.

Logistic Regression

- Logistic regression is based on the concept of Maximum Likelihood estimation. According to this estimation, the observed data should be most probable.
- The weighted sum of inputs is fed through an activation function that can map values in between 0 and 1.
- Such activation function is known as **sigmoid function**, sigmoid curve or S-curve.



References

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